

House Price Prediction Using Machine Learning

Objective:

The objective of this project was to develop a machine learning model capable of predicting house prices based on various property features such as area, number of bedrooms, bathrooms, stories, parking facilities, and furnishing status.

Dataset Used:

The Housing Prices Dataset obtained from Kaggle was used for this project. The dataset contains information about various residential properties and their corresponding market prices.

Data Preprocessing:

The dataset was loaded using Pandas and inspected for missing values and duplicate records. Categorical variables were converted into numerical form using one-hot encoding. This preprocessing step ensured compatibility with machine learning algorithms.

Models Used:

Two machine learning models were trained and evaluated:

- Linear Regression
- Random Forest Regressor

The dataset was divided into training and testing sets using an 80:20 ratio.

Results:

Linear Regression

MAE = 970043.4039201641

RMSE = 1324506.9600914386

R2 = 0.6529242642153184

Random Forest

MAE = 1021546.0353211008

RMSE = 1400565.9728553821

R2 = 0.611918531405699

Key Findings:

The analysis revealed that area, bathrooms, bedrooms, parking spaces, and number of stories had the strongest influence on house prices. Properties with larger areas and better amenities generally had higher market values.

Conclusion:

Among the two models tested, the Random Forest Regressor achieved better performance and provided more accurate predictions. Machine learning can significantly improve property valuation and assist buyers, sellers, and real estate companies in making informed decisions.